SOUMYADYUTI DEY

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SUMMARY

Software Engineer and BCA graduate with hands-on experience in **Computer Vision**, including real-time **pose estimation**, geometric movement analysis, and **multimodal video understanding**. Professional experience developing **AI/ML applications, Python services, and cloud-deployed systems** at early-stage companies. Submitted open-source contributions to **Kornia and Kornia-SLAM**, spanning 2D/3D geometry and keyframe management. Seeking research opportunities in **Visual SLAM, 3D perception, and geometric vision**, with a focus on reproducible experiments, benchmarking, and accuracy–efficiency trade-offs.

EDUCATION

Bachelor of Computer Application, The NorthCap University 2023 – 2026
Relevant Coursework: AI/ML, Cloud Computing, NLP, Cybersecurity **CGPA: 8.66/10**

SKILLS

- **Computer Vision & Imaging:** OpenCV, MediaPipe Pose, Kornia, 2D landmarks, pose geometry, FFmpeg
- **Machine Learning & Tools:** Python, NumPy, PyTorch, TensorFlow, Hugging Face, Git/GitHub, Linux
- **Backend & Data:** FastAPI, Django, PostgreSQL, MySQL, MongoDB, REST APIs, GraphQL
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD, Terraform
- **Additional Engineering:** JavaScript, TypeScript, Go, React, Next.js, Node.js

OPEN-SOURCE CONTRIBUTIONS

Kornia-SLAM – Keyframe Management (PR #170) *[Under review]*
Proposed a refactor of keyframe insertion and culling policies, including frame-based/parallax insertion and configurable redundancy culling, while preserving default behavior and adding tests.

Kornia Core – 2D/3D Geometry (PR #6554) *[Under review]*
Proposed empty-batch handling for 2D/3D box-cropping operations and regression tests to preserve existing behavior and API semantics.

EXPERIENCE

Full Stack Developer – AI/ML Jan 2026 – Jun 2026
Tynari *Remote / Hyderabad, India*

- Architected and maintained full-stack applications at Printeast, integrating a React frontend with Node.js/Python services and PostgreSQL/MongoDB for an AI-driven print-on-demand platform.
- Integrated and deployed AI/ML models using PyTorch/TensorFlow and external APIs (OpenAI, Hugging Face), building intelligent modules for content generation, recommendations, and analytics; optimized services and deployed on AWS.

Software Development Engineer Oct 2025 – Dec 2025
Yume *Remote / Hyderabad, India*

- Architected a scalable City Data Pipeline using PostgreSQL and GraphQL, establishing a unified data source for web (Next.js) and mobile (React Native) applications.
- Engineered a Python-based data ingestion engine with staging-to-production validation, automating the processing and cloud storage (AWS S3) of thousands of travel data points.

PROJECTS

Asansync – AI-Powered Yoga Coaching: Built a real-time pose-analysis prototype using **OpenCV, MediaPipe Pose, and NumPy**; extracted **2D body landmarks**, computed **joint-angle and reference-pose similarity scores**, and displayed skeleton overlays with corrective feedback for yoga/exercise movements.

UnReel – Multimodal Video Intelligence: Developed an **8-stage pipeline** using **FFmpeg** frame extraction, **Whisper** transcription, **Gemini** vision-language analysis, and **RAG** to interpret visual scenes, identify landmarks, and connect visual evidence with transcript context; reduced manual analysis time by **40%** in validation runs.

Scribe – AI Note-Taking App: Built PDF text extraction and **Tesseract-based handwritten-note OCR** to digitize scanned study notes, combining **RAG-based summarization**, contextual quiz generation, and mind maps (**FastAPI, LangChain**); improved backend inference latency by **30%**; won **SeamEdu 2025 Best Performing Web Application**.

CERTIFICATIONS

AWS Certified Developer – Associate (2024)
Oracle Cloud Infrastructure Foundations – Associate (2025)